COMPUTABILITY / DIOPHANTINE CONSISTENCY

A direct integer equation for the consistency of ZFC

Joint Pell certificates and exact integral elimination

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25 October 2026

Principal delivered equation

$$P_{85}(y_0, \dots, y_{84}) = \sum_{j=0}^{23} r_j(\boldsymbol{y})^2 = 0$$

85 unrestricted integer unknowns; exact degree 16,964;
1,788 live binary arithmetic operations.

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{y} \in \mathbb{Z}^{85}\ P_{85}(\boldsymbol{y}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This statement uses the explicitly identified inherited C222/ZF/ZFC calibration. The new arithmetic composition and coordinate eliminations have ordinary mathematical proofs and exact finite certificates. The appendices specify every coefficient and substitution of the principal polynomial. No choice of coefficient depends on the truth of consistency.

Research status. AI-assisted research with separately assigned internal mathematical and artifact reviews. This is a concrete improvement within the inspected direct-calculus family, not a global simplicity minimum or a claim of worldwide priority. The archive also contains alternative variable/degree tradeoffs. It supplies no new end-to-end Lean or Rocq compilation and proves neither consistency nor inconsistency.

Abstract

We construct a fixed polynomial over the integers whose integer unsolvability is equivalent, provably in ZFC, to syntactic consistency of ZFC. The principal polynomial has 85 unrestricted integer unknowns, exact total degree 16,964, and a division-free evaluator with 1,788 live binary operations. Its 24 residual squares and all polynomial reconstruction maps are specified explicitly. The construction combines the cofinal CRT anchor of ProveIt package 173 with the joint signed-index and power certificate of package 176. Eleven old certificate coordinates are replaced by one; 124 further coordinates are eliminated using acyclic equations with unit coefficients. Each of these 124 coordinate reconstructions is an integer polynomial. The new composition retains the numerical C222 proof calculus and has no K215 simulation, Jones polynomial, or large encoded-machine coefficient tuple. We separate the unbounded number-theoretic and logical proof from finite algebra verification, give exact degree and variable-dependence certificates, and record the larger literal coefficients introduced by the CRT-anchor parent. The result improves selected inspected direct variable/degree/circuit bounds; equations in fewer variables already exist in other regimes. A restricted investigation of additional Pell centers is included to explain why a plausible competing route did not improve its parent degree.

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1 Result, scope, and comparison

The task is to give one actual polynomial, with all of its integer coefficients determined, for which ZFC proves an equivalence between integer unsolvability and its own syntactic consistency. Merely invoking the existence part of the DPRM theorem would not specify such a polynomial. A bounded family of consistency tests would also not meet the task. This report gives a finite, independently replayable polynomial definition and identifies the precise inherited theorem that supplies its logical meaning.

There is no single established ordering of "simplest." Number of unknowns, degree, coefficient height, description length, evaluation cost, and proof complexity can conflict. Our principal choice gives a substantial improvement of an inspected direct proof-calculus family. It is not the fewest-variable representation in the entire repository: the repository already contains much smaller variable counts in different degree and coefficient regimes. Literature on universal pairs likewise treats variable count and degree jointly [6].

Theorem 1.1 (Principal consistency equation)**.** *Let $P_{85} \in \mathbb{Z}[y_0, \ldots, y_{84}]$ be the polynomial specified in Sections 3 and 5 and Appendices A–C. Using the inherited calibration in Section 7, ZFC proves*

$$\exists \boldsymbol{y} \in \mathbb{Z}^{85}\ P_{85}(\boldsymbol{y}) = 0 \quad \longleftrightarrow \quad \exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

Consequently ZFC proves

$$\neg\exists \boldsymbol{y} \in \mathbb{Z}^{85}\ P_{85}(\boldsymbol{y}) = 0 \quad \longleftrightarrow \quad \mathrm{Con}(\mathrm{ZFC}).$$

The polynomial has exact total degree 16 964*, depends on each of its 85 variables, and is a sum of 24 polynomial squares. Its supplied division-free evaluator has 1,788 live binary additions, subtractions, and multiplications.*

The proof concerns the standard syntactic statement: no finite proof of falsity in a fixed effective presentation of ZFC. It does not assume consistency, infer the truth of arbitrary ZFC theorems, or require a transitive model. All occurrences of "integer" in the delivered equations mean an unrestricted integer. Positivity assertions below are conclusions of retained equations or hypotheses of stated lemmas.

1.1 Measured alternatives

Table 1: Delivered integer-polynomial tradeoffs. All coordinates range over $\mathbb{Z}$.

Variables	Exact degree	Residuals	Operations	H
84	2 097 732	23	1 775	7 342 557
85	**16 964**	**24**	**1 788**	**59 805**
86	1 092	25	1 797	5 341
87	956	26	1 804	5 093
88	444	27	1 807	2 149
90	256	29	1 819	1 297
100	64	39	1 872	390
109	32	48	1 895	308

All degrees in this table are exact, not just estimates propagated through a circuit. "Operations" counts live binary $+, -, \times$ instructions; a square and multiplication by a fixed integer each cost one operation. This is an algebraic operation count, not a bit-cost bound. The coefficient exponent H certifies $\max_\alpha |[\boldsymbol{y}^\alpha]P| \le 2^H$; it is an upper bound on expanded coefficients, not an exact height.

The comparison snapshot is ProveIt commit `ecb9a4b4dda617512952525d50f17343700aac52`. Package 177 reports a direct 87-integer polynomial of degree 8 388 612 with 2,447 operations, a 94-integer polynomial of degree 256 with 2,468 operations, a 104-integer polynomial of degree 64 with 2,451 operations, and a 111-integer polynomial of degree 32 with 2,502 operations [3]. Our 85-, 90-, 100-, and 109-integer profiles improve the corresponding inspected dimension/degree/circuit comparisons.

That statement has an explicit cost qualification. The largest absolute integer literal in the new supplied circuits is $1\,099\,511\,627\,780 = 2^{40} + 4$; package 177's corresponding direct circuits use literals of magnitude at most 256. Large literal coefficients arise in the inherited frame because the first requested power has index $2^{20}t$. Thus the new equations do not dominate every coefficient-sensitive simplicity measure. Neither actual expanded term counts nor exact expanded coefficient heights are claimed for the high-degree equations.

Two ZIPs were present in `docs/incoming` at the snapshot: `ConZFC_Mixed_Base_Pooling_2026-10-25.zip` and `ConZFC_Affine_Horizons_Centered_Certificates_2026-10-26.zip`. Their binary contents were not accessible through the available repository reader. They are outside the novelty comparison. The mathematical theorem and literal equations do not depend on a novelty claim about these unread submissions.

1.2 What is new

The generic ingredients are credited rather than renamed as new theorems. Package 173 supplies the cofinal CRT index used as a Pell anchor and the affine growth horizon. Package 176 proves joint authentication of signed indices and powers. Acyclic elimination with unit coefficients already appears in the earlier direct constructions, including package 152. The contribution here is their explicit compatible composition, the source-specific interface proof, and the resulting frozen integer polynomials and certificates. In particular, the 147 retained quadratic rows, all sixteen power requests, and the actual signed frame are checked together; the composition is not justified by matching variable counts alone.

2 The calibrated arithmetic source

The immediate source is package 173's `source/anchor219/equation.json`, preserved as identical literal polynomial data in `source/anchor219.json`. The working copy has one additional trailing newline; the exact upstream bytes and their authenticated Git blob identity are retained separately. It has 219 integer coordinates $v_0, \dots, v_{218}$ and 158 quadratic residuals $R_0, \dots, R_{157}$. The source profile enforces, in its arithmetic interpretation,

$$t = 16 + 4d^2, \qquad K = 2^{20}t, \qquad F = t^K, \qquad C = 2v_{16}.$$

These variable powers describe the interpretation of the polynomial certificate. They are not operations in the final polynomial. In the literal source $t = v_0$, $F = v_2$, and the shared Pell parameter is $A = v_{137} + 1$.

The source binds a numerical proof calculus directly to arithmetic. Finite derivations are represented by local constructor and rule conditions; a generalized binomial/CRT encoding enforces these local conditions, and a Horn-flow argument recovers a finite refutation. There is no intermediate Turing-machine computation in this branch. The calibration is inherited, and its boundaries are stated in Section 7.

2.1 Why the affine growth index is sufficient

The deleted derived-label range of package 173 is replaced by a uniform constructor bound. Its finite constructor-depth ledger uses $\lambda = 2^{20}$ and gives local positive labels below $t^{\lambda-1}$. Write $L = t^{\lambda}$. The growth request $F = t^{\lambda t}$ then gives $F = L^t$. The boundary case matters: an argument requiring $F > L^t$ would not suffice for this source.

The relevant large-evaluation lemma says that, for positive labels at most L, $L \geq t$, and $F \geq L^t$, equality of two products of factors $1 + Fj$, with at most t factors on one side and at most $2t + 1$ on the other, forces equality of their multisets of labels. First compare the number of factors: an extra factor produces a value exceeding the other side because $F + 1 > L^t$. Once both sides have $m \leq t$ factors, their elementary symmetric coefficients lie in $[1, F]$ in positive degrees. Differences lie strictly between $-F$ and F. Reduction at the least nonzero coefficient modulo F forbids a nonzero difference polynomial from vanishing at F. Factorization over $\mathbb{Q}$ then recovers the labels. Package 173 supplies the full boundary-valid proof and the constructor-depth certificate [1].

The other useful source change is cofinality. The actual CRT index C can be chosen positive, even, and above all other requested indices inside its prescribed residue class. The CRT modulus is odd, so the allowed even representatives form an unbounded progression. The radix and Pell frame are chosen afterward. Consequently C can serve as the semantic anchor without an unjustified bound on a previously fixed representative. This ordering is retained unchanged here.

3 A literal joint polynomial certificate

Delete the following source coordinates and rows:

$$D_v = \{144, 145, 146, 147, 148, 149, 204, 214, 215, 216, 217\},$$
$$D_R = \{102, 103, 104, 105, 106, 107, 147, 153, 154, 155, 156\}.$$

Every coordinate in D_v occurs only in rows in D_R. Retain the other 208 coordinates and append $v_{219} = z$. The resulting 209-coordinate system keeps the 147 rows R_j for $j \notin D_R$ and replaces all eleven deleted rows by the single residual defined below.

Let the five bases be

$$n_0 = v_0, \quad n_1 = v_3, \quad n_2 = v_{189}, \quad n_3 = v_{100} - 1, \quad n_4 = v_{100}.$$

For each row of Table 2, m_i is the requested output, k_i the requested exponent, and (D_i, Y_i, Z_i) the stored Pell data. The retained gates give $Z_i = AY_i$, and the actual abscissa is $X_i = D_i + Z_i$. Affine shifts in the table are part of the exact definition; they must not be dropped.

There are sixteen requests and fourteen distinct principal pairs. Requests sharing the same (D_i, Y_i, Z_i) also have the same k_i. Use one copy of each distinct pair in the following sum:

$$\Omega = \sum_{\text{distinct pairs } i} \big((X_i - 1)^2 + (Y_i - k_i)^2\big).$$

For $0 \leq j \leq 4$, let I_j be the set of requests in base group j, and define ordinary polynomial abbreviations

$$d_A = A - 1, \qquad T_j = 2An_j - n_j^2 - 1, \tag{1}$$
$$H_j = \sum_{i \in I_j} Y_i(m_i - X_i), \qquad S_j = \sum_{i \in I_j} \big((m_i - D_i)^2 - Y_i^2\big), \tag{2}$$
$$E_j = S_j - 2n_j H_j, \qquad \Pi = \prod_{j=0}^{4} T_j. \tag{3}$$

Table 2: The sixteen literal power requests. The base in group j is n_j.

i	Group j	m_i	k_i	D_i	Y_i	Z_i
0	0	v_2	$2^{20}v_0$	v_{186}	v_{187}	v_{188}
1	1	v_{17}	v_1	v_{177}	v_{178}	$v_{179}+1$
2	2	$v_{100}-1$	v_1	v_{177}	v_{178}	$v_{179}+1$
3	4	v_{101}	v_0+v_2	v_{150}	v_{151}	v_{152}
4	3	v_{102}	v_0	v_{141}	v_{142}	v_{143}
5	4	v_{106}	v_8-1	v_{153}	$v_{154}-1$	v_{155}
6	4	v_{109}	v_9	v_{156}	v_{157}	v_{158}
7	4	v_{112}	v_{10}	v_{159}	v_{160}	v_{161}
8	4	v_{115}	v_{11}	v_{162}	v_{163}	v_{164}
9	4	v_{118}	v_{12}	v_{165}	v_{166}	v_{167}
10	4	v_{122}	v_{13}	v_{171}	v_{172}	v_{173}
11	4	v_{125}	v_{14}	v_{174}	v_{175}	v_{176}
12	4	v_{130}	v_{15}	v_{180}	v_{181}	v_{182}
13	3	v_{119}	v_4	v_{168}	v_{169}	v_{170}
14	3	v_{126}	v_1	v_{177}	v_{178}	$v_{179}+1$
15	4	v_{133}	$2v_{16}$	v_{183}	v_{184}	v_{185}

The joint residual is the following polynomial, without a division:

$$\boxed{\mathcal{J} = \Omega\Pi + d_A \sum_{j=0}^{4} n_j E_j \prod_{\ell \neq j} T_\ell - z\, d_A \Pi.} \tag{4}$$

All sums and products have the fixed finite index sets just printed. They introduce no quantified variables. The intermediate polynomial is

$$P_{209} = \mathcal{J}^2 + \sum_{j \in \{0,\ldots,157\} \setminus D_R} R_j^2. \tag{5}$$

It has 148 residual squares and degree at most 26.

4 Why orientation and powers can share one certificate

This section supplies the unbounded argument, including its order of dependence. It is essential to classify the raw Pell solutions before identifying their indices with the requested exponents.

4.1 Information available before the new residual is used

The retained signed frame and squared-anchor comparison equations give

$$W \geq 258, \qquad |A| > W^W, \qquad k_i^2, n_j^2, m_i^2, C^2 < W.$$

Set $b = \lfloor \sqrt{W-1} \rfloor$, so $b \geq 16$. The comparison congruences identify the anchor's positive classified index as $|C|$. The squared-ordinate bound then bounds the absolute classified indices of all principal pairs by $|C| \leq b$. These conclusions use neither the removed orientation row nor the removed power rows.

Let $\mathsf{X}_n(A), \mathsf{Y}_n(A)$ denote the signed Pell sequences. Every raw principal pair can therefore be written

$$(X_i, Y_i) = \epsilon_i(\mathsf{X}_{\nu_i}(A), \mathsf{Y}_{\nu_i}(A)), \quad \epsilon_i \in \{-1, 1\}, \quad \nu_i \in \mathbb{Z}, \quad |\nu_i| \le b. \tag{6}$$

At index zero this includes both $(1, 0)$ and $(-1, 0)$. For negative A, the signed recurrence handles the parity signs; no positivity of every abscissa is imposed.

4.2 The small integer with a common sign

Modulo $d_A = A - 1$, the canonical recurrences give $\mathsf{X}_\nu(A) \equiv 1$ and $\mathsf{Y}_\nu(A) \equiv \nu$. Consequently

$$\Omega \equiv o := \sum_i ((\epsilon_i - 1)^2 + (\epsilon_i \nu_i - k_i)^2) \pmod{d_A}, \qquad o \ge 0. \tag{7}$$

At a nonzero base, $\gcd(n_j, T_j) = 1$ because $T_j \equiv -1 \pmod{n_j}$. The Pell recurrence reduces modulo T_j to

$$X_i + (n_j - A)Y_i \equiv \epsilon_i n_j^{\nu_i} \pmod{T_j}.$$

Negative exponents in this proof-only congruence are defined by the modular inverse of n_j. Define $\delta_i = m_i - X_i + (A - n_j)Y_i = m_i - D_i - n_j Y_i$. A direct polynomial calculation, using $Z_i = AY_i$, gives

$$\sum_{i \in I_j} \delta_i^2 = E_j - T_j \sum_{i \in I_j} Y_i^2. \tag{8}$$

All denominators in the argument are nonzero. The number d_A has the sign of A and $|d_A| \ge W^W$. If $n_j \ne 0$, then $|T_j| > W^W$ and n_j/T_j has the sign of A. If $n_j = 0$, then $T_j = -1$, and the weighted summand vanishes. Dividing (4) and applying (8) shows that

$$\frac{\Omega}{d_A} + \sum_j \sum_{i \in I_j} \frac{n_j \delta_i^2}{T_j} \quad \text{is an integer.}$$

Put $C_* = \prod_{j : n_j \ne 0} n_j^b$, a nonzero integer. For a request in a nonzero group define

$$e_i = C_* m_i - \epsilon_i \frac{C_*}{n_j^b} n_j^{b + \nu_i}.$$

Here C_*/n_j^b is an integer and $b + \nu_i \ge 0$; thus this definition uses ordinary integer powers in the proof. We have $C_* \delta_i \equiv e_i \pmod{T_j}$. Reducing each numerator in its own denominator proves that

$$V_* = \frac{C_*^2 o}{d_A} + \sum_{j : n_j \ne 0} \sum_{i \in I_j} \frac{n_j e_i^2}{T_j} \quad \text{is an integer.} \tag{9}$$

No coprimality between different denominators is required. This is particularly useful when an index modulus and a power modulus have large common factors.

Every nonzero summand of (9) has the same sign. The conservative bounds of the joint-certificate theorem accommodate at most fifteen pairs and seventeen requests; our fourteen and sixteen are within these bounds:

$$o \le 120 b^2, \qquad |C_*| \le b^{5b},$$
$$C_*^2 o + \sum_{j,i} |n_j| e_i^2 \le 120 b^{10b+2} + 68 b^{12b+1} \le 188 b^{12b+2} < W^{6b+2} \le W^W.$$

For completeness, $|e_i| \le 2b^{6b}$ follows from $|m_i| \le b$, $|C_*| \le b^{5b}$, and $|n_j^{\nu_i}| \le b^b$ when $\nu_i \ge 0$; negative indices only decrease that absolute-value bound. There are at most seventeen terms, each carrying a factor $|n_j| \le b$. The strict comparison uses $188 < W$, $b^2 < W$, and $6b + 2 \le b^2 + 1 \le W$ for $b \ge 16$.

It follows that $|V_*| < 1$. Since V_* is an integer, it is zero. The common-sign property then forces $o = 0$ and every $e_i = 0$. Equation (7) gives $\epsilon_i = 1$ and $\nu_i = k_i$ for each pair, and $e_i = 0$ gives the exact power $m_i = n_j^{k_i}$ at every nonzero base. When $|n_j| \ge 2$, an integer value of this power excludes a negative k_i.

Proposition 4.1 (Joint authentication)**.** *Under the retained frame, norm, gate, and squared-anchor hypotheses, $\mathcal{J} = 0$ simultaneously authenticates the canonical signs and indices of all fourteen pairs and the sixteen powers at every nonzero base. Conversely, canonical pairs and exact powers admit an integer z making $\mathcal{J} = 0$.*

Proof. Soundness was proved above. For completeness, the canonical base-one congruences give $d_A \mid \Omega$. Exact powers and the modular Pell identity give $T_j \mid \delta_i$, hence $T_j \mid E_j$ by (8). Therefore

$$z = \frac{\Omega}{d_A} + \sum_{j : n_j \neq 0} \frac{n_j E_j}{T_j}$$

is an integer satisfying (4). Zero-base contributions are zero. These divisions construct witnesses in the proof; none occurs in the defining polynomial. □

4.3 Direct restoration, including zero bases

The new source need not repeat the parent's longer arithmetic decoding argument. It suffices to restore the eleven deleted source coordinates. For a nonzero base, joint authentication gives the canonical congruence and the exact requested power, so $T_j \mid \delta_i$ and $T_j \mid E_j$. For a zero base, $T_j = -1$, so both divisibilities hold automatically. Thus the temporary zero-base branches create no restoration gap.

Restore the three removed singleton moduli as T_j and their quotients as $-\delta_i/T_j$. For either old pooled base group restore

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

The preceding argument proves these are integers and makes both old pooled rows zero. Restore the orientation quotient as Ω/d_A. Source row 147 differs from $\Omega - d_A v_{204}$ by minus the sum of the fourteen retained norm rows; this exact identity is included in the independent source audit. Thus every P_{209} zero extends to a source219 zero. Conversely, the old source rows themselves establish the divisibilities needed to construct z in the displayed completeness formula. Every source219 zero therefore projects to a P_{209} zero. No claim of an integer restoration away from the zero sets is needed here.

5 Eliminating 124 coordinates over the integers

The preceding reduction is semantic: its quotient witnesses are restored only after proving divisibility. The next reduction is purely polynomial and has an integer reconstruction for every free input.

Lemma 5.1 (Acyclic unit elimination)**.** *Let an integer polynomial system contain selected rows $\epsilon_v v + F_v = 0$, where $\epsilon_v \in \{-1, 1\}$ and F_v does not contain v. Suppose the dependence graph among selected variables, with v depending on those selected variables occurring in F_v, is acyclic. Assign arbitrary integers to the unselected variables and recursively set $v = -\epsilon_v F_v$. This defines an integer polynomial map Φ from the free coordinates to all coordinates. The remaining equations after substitution have an integer zero if and only if the original system does. Projection and Φ are inverse on the two zero sets.*

Proof. Topologically order the selected variables with dependencies first. At each step the displayed substitution is an integer polynomial in already defined coordinates. It makes its selected row zero identically. Any zero of the original system must have exactly these coordinate values, by induction along the same order. The remaining rows then vanish on either side precisely when their substituted versions vanish. This proves both directions and the inverse claim. □

The unit condition is crucial. Solving $2v = u$ over the rationals and discarding the equation would lose the parity condition on u. Solving $av = F$ without a divisibility certificate can introduce spurious integer inputs. Our 124 selected coefficients are all literal 1 or -1, and no selected variable occurs elsewhere in its defining row.

Appendix A lists the 85 free source indices and Appendix B gives every selected row, variable, and unit coefficient in dependency order. Let Φ be the resulting map. The 23 unselected old rows are

$$\begin{aligned}\mathcal{K} = \{&1, 42, 83, 99, 101, 109, 111, 113, 115, 117, 119, 121, 123,\\ &125, 127, 129, 131, 133, 138, 140, 149, 150, 151\}.\end{aligned} \tag{10}$$

The principal equation, now fully specified, is

$$\boxed{P_{85}(\boldsymbol{y}) = \mathcal{J}(\Phi(\boldsymbol{y}))^2 + \sum_{j\in\mathcal{K}} R_j(\Phi(\boldsymbol{y}))^2 = 0.} \tag{11}$$

The fixed finite source row table, request table, free-index list, and pivot table determine every coefficient of this polynomial. They are one finite definition, not a program that searches for a suitable polynomial or assumes an unspecified universal parameter. Source coordinates outside the free list are polynomial abbreviations, not additional unknowns.

Since selected rows pull back identically to zero, the exact polynomial identity $P_{85} = P_{209} \circ \Phi$ also holds. The reverse map on zero sets is coordinate projection. This makes the existential equivalence particularly straightforward to formalize in ZFC.

6 Exact complexity certificates

6.1 Degree is proved, including noncancellation

An arithmetic DAG propagates a valid upper degree bound by taking a maximum at additions and adding degrees at multiplications. Such a bound alone is not an exact-degree proof: leading terms can cancel. For the principal equation the retained residual degree bounds are

$$(6, 4096, 8482, 50, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 14, 12, 2, 2, 8, 39).$$

Their order is the increasing list $\mathcal{K}$, followed by $\mathcal{J}$. The maximal residual is source row 83, of degree at most 8,482 after substitution.

For a completely exact lower bound, put $p = 1\,000\,000\,007$ and let $s_0 < \cdots < s_{84}$ be the free source indices. Evaluate the propagated top homogeneous component on the integer direction $a_i = s_i + 3$. Its residue at the row-83 residual is

$$[\lambda^{8482}]R_{83}(\Phi(\lambda\boldsymbol{a})) \equiv 140\,621\,492 \not\equiv 0 \pmod{p}.$$

The leading term has a simple source-level explanation. In row 83, $-v_{79}v_{95}$ is the unique highest-degree term after reconstruction. The reconstructed coordinates have exact degrees 8,198 and 284; their top

components on the same ray are respectively 46 952 950 and 64 682 962 modulo p. Every other term of row 83 has degree at most 12. Their product gives the displayed nonzero residue. This modular calculation is an exact finite witness that the integer coefficient is nonzero; it is not a probabilistic polynomial-identity test. Therefore that residual has exact degree 8,482. Over the real numbers a sum of polynomial squares cannot cancel its maximal homogeneous components. Equivalently, on this integer ray, the leading real coefficient of the sum is a sum of squares with a nonzero summand. Hence $\deg P_{85} = 2 \cdot 8482 = 16\,964$. The last step is over $\mathbb{R}$ or $\mathbb{Q}$, not a claim that sums of squares cannot cancel modulo p.

6.2 Every counted variable actually occurs

A syntactically live input need not affect the resulting polynomial; the expression $x - x$ demonstrates the distinction. The archive gives one exact modular point and all 85 partial derivative values computed by reverse differentiation of the literal DAG. Every value is nonzero. Thus each integer polynomial $\partial P_{85}/\partial y_i$ is nonzero, and all 85 coordinates genuinely occur. The point was found by a deterministic search, but verification of the supplied witness uses only exact integer arithmetic modulo the displayed modulus.

6.3 Coefficient and evaluator accounting

The checker propagates upper bounds on the coefficient ℓ^1 norm. For integer polynomials f, g,

$$\|f \pm g\|_1 \le \|f\|_1 + \|g\|_1, \qquad \|fg\|_1 \le \|f\|_1 \|g\|_1.$$

Bounding each norm by a power of two and adding the residual squares gives $\max_\alpha |[\boldsymbol{y}^\alpha] P_{85}| \le 2^{59805}$. No huge expanded polynomial needs to be materialized to prove this bound. The largest circuit literal $2^{40} + 4$ and the expanded-height bound are deliberately reported separately.

The JSON specification retains some nodes solely for reconstruction and certificate comparison. The operation count includes only binary operations reachable from the final polynomial output. The principal file contains 3,559 stored nodes, while its live evaluator uses 1,788 binary operations. Constants and input nodes are not counted as binary operations. These conventions match the stated comparison metric and do not assert optimal arithmetic-circuit size.

7 Logical calibration and internal ZFC meaning

The inherited chain is

$$\begin{gathered}\text{square-shell C222 refutation} \iff \text{original C222 derivation of code 1},\\ \text{original C222 derivation of code 1} \iff \text{a finite ZF refutation},\\ \text{a finite ZF refutation} \iff \text{a finite ZFC refutation}.\end{gathered}$$

The named Lean source interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The effective first-order axiom predicate is the repository's `ZFCax_s`, with finite proof relation `BProv` and falsity `Form.fBot`. The square-shell syntax change is an ordinary effective recoding; it is not a literal identification of two differently coded formula constructors.

The source calculus-to-ZF bridge is a substantial inherited theorem. Its model arguments use the internally well-founded part of a quotient construction, and the ZF-to-ZFC consistency transport uses the constructible model. None of this is replaced by assuming that a consistent theory has an externally

well-founded or transitive model. The present packet identifies and retains the relevant source proofs; it does not freshly compile the full transitive import closure. The source audit and provenance ledger should be read as such, not as a new kernel-checked theorem about the entire development.

For the new steps, the formalizability claim is direct. Pell recurrences, their classification and congruences, finite CRT, polynomial substitution, prime factorization, and induction over finite syntactic objects are ordinary constructions provable in ZFC. The inequalities in Section 4 are uniform in the internally coded integers. They do not quantify only over externally standard proof codes. In a nonstandard model of ZFC, the theorem concerns that model's integers and finite derivations, interpreted by the same internally proved lemmas.

Combining the inherited source theorem, the parent-restoration equivalence, and Lemma 5.1 proves Theorem 1.1. Negating the existential refutation statement gives exactly the usual arithmetized Con(ZFC). It gives no independent proof that P_{85} has no integer zero.

8 Verification, reproduction, and integration

The archive is self-contained for reproducing the new finite polynomial checks. Python 3.10 or later and its standard library suffice; a TeX installation is needed only to rebuild this PDF. From the extracted package root, run

```
python3 -B verify.py
python3 -O -B verify.py
```

The verifier checks the source hashes, exact retained/deleted row interface, the joint pool identities, unit pivots, acyclicity, complete substitution of source DAG nodes, the sum of squares, exact degree witnesses, all-variable dependence witnesses, and reported finite metrics. Deliberate corruptions are required to fail. It imports neither the projection producer nor its search routine. Optimization-mode replay checks that substantive gates are not implemented merely as removable Python assertions.

The source and new proofs establish unbounded facts. The finite verifier authenticates their literal algebraic instances. Passing checks does not, by itself, establish the Pell classification, logical calibration, or absence of solutions. No search for a Diophantine zero is performed or needed.

The principal JSON contains a topologically ordered list of constants, variables, and binary operations; its final output is the actual integer polynomial. A supplied evaluator can inspect its input order, evaluate it exactly, or evaluate it modulo a chosen positive integer. A modular zero would not constitute an integer solution. Evaluating the high-degree forms exactly can produce large values even at small inputs.

For repository intake, add the ZIP as an independent manuscript package in `docs/incoming`, then place it according to the repository's intake procedure. Preserve the source ledger, inherited notices, proof-boundary statement, and domain distinction. The package does not replace the maintained fifteen-positive-variable display, claim an improvement in every simplicity measure, or silently upgrade an ordinary proof to a fresh Lean/Rocq build.

9 An unsuccessful route worth recording

We also examined the latest ten-positive-variable degree-208 construction of package 178 [4]. Its center approximates a square root of a product of Pell expressions using polynomial quotients at infinity. A

natural extension is to retain more terms in

$$\sqrt{A^2-1} = A - \frac{1}{2A} - \frac{1}{8A^3} - \frac{1}{16A^5} - \cdots .$$

The degree relevant to the final equation is a weighted total degree, not just the degree in A. Taking the full polynomial part in A does not improve the parent's degree: a term involving high powers of Y survives with the same weighted degree.

In the actual variables $A = XY + Y + 1$, with weights $(X, Y, k, U, b, c) = (2, 2, 1, 7, 4, 3)$, the first nonpolynomial term of the relevant scale-four root expansion is $3X^{-1}Y^7k^4U^3b$, of weight 41. The next terms are $-3X^{-2}Y^7k^4U^3b + 12X^{-1}Y^6k^4U^3b$, of weight 39. Multiplication by X permits the first correction but leaves a defect of weight 108, compared with the parent's 106; its final degree is worse. Multiplication by $X + 1$ can cancel one further term but leaves the same defect weight 106. The independently checked notes in `research/` retain the exact restricted calculation and its assumptions.

A stronger restricted statement follows from these coefficients. Let

$$A = XY + Y + 1, \quad E = 1 + XY^2k^2 + X^2Y^4k^2, \quad \Delta = A^2 - 1,$$
$$G = E + \Delta U^2E^2, \qquad V = (1 + 2\Delta U^2E)b - c.$$

Proposition 9.1 (Affine-scale obstruction)**.** *In $\mathbb{Q}[X, Y, k, U, b, c]$ with the weights above, for every nonzero $r = aX + b_0Y + d \in \mathbb{Q}[X, Y]$ and every polynomial C,*

$$\deg_w(C^2 - 16r^2GV^2) \geq 106.$$

Proof. Introduce a weighted Laurent parameter and choose the formal root $S = 4\sqrt{G}\,V$ with leading monomial $8X^7Y^{11}k^4U^3b$ of weight 65. Above weight 41 its components are polynomials. Write $K = k^4U^3b$. Its nonpolynomial components at weights 41, 39, and 37 are exactly

$$\frac{3KY^7}{X}, \qquad -\frac{3KY^7}{X^2} + \frac{12KY^6}{X}, \qquad \frac{3KY^7}{X^3} - \frac{15KY^6}{X^2} + \frac{15KY^5}{X}.$$

They are obtained by recursively squaring and comparing coefficients; the supplied standard-library checker verifies these finite identities with rational coefficients. For nonzero constant r, a defect below 106 requires agreement of C with one sign of rS through weight 41, which its first nonpolynomial term prevents. For nonconstant r, the leading weight is 67, so a defect below 106 requires agreement through weight 39. At weight 43 the nonpolynomial part is $3b_0KY^8/X$, forcing $b_0 = 0$. At weight 41 it is $3(d - a)KY^7/X$, forcing $d = a$. The remaining nonzero scale is $a(X + 1)$, whose weight-39 nonpolynomial part is $-3aKY^6/X$. This is impossible for a polynomial center. The degree conclusion uses $(C - rS)(C + rS)$: after choosing the leading sign, the second factor retains weight 65 or 67, so its product with the first unmatched coefficient cannot cancel in the rational function field. □

This is a theorem about the displayed abstract polynomial ring and affine scales. It is not a global lower bound for consistency equations, arbitrary nonlinear scales, or different encodings. In particular, it prevents recycling this specific unsuccessful improvement attempt without claiming that degree 208 is universally minimal.

10 Further research questions

1. Can the joint orientation-and-power residual be applied before more of the source's quadratic coordinates are introduced? The present composition saves certificate variables and then projects; a compiler that preserves this sharing from the outset might reduce both source size and literal coefficient growth.

2. Can a smaller constructor majorant replace 2^{20} while preserving the boundary-valid multiset-separation proof? This targets the main coefficient tradeoff rather than just the variable count. An improved bound must cover every local constructor on every allowed parameter tuple, not only sampled syntax.
3. Which dependency structures prevent the remaining unit eliminations? A certified search over integer coordinate shears, followed by acyclic unit pivots, could expose equations that are solvable integrally but not monic in the current coordinates. Any nonunit pivot requires an explicit congruence or divisibility argument.
4. Can the new low-dimensional forms be given sharper expanded coefficient bounds, or smaller evaluation circuits, without changing their zero sets? Circuit simplification should preserve an exact map to the frozen definition and recount all live constant products.
5. Can the new source-interface theorem and reconstruction map be formalized on top of the existing ZFC endpoint? A modest first target is the 209-to-85 polynomial pullback and zero-set equivalence; the larger target is an explicit object-language ZFC derivation of the full consistency equivalence.
6. Can a different complete proof calculus improve the entire description, rather than its arithmetic certificate alone? The comparison should charge syntax, coefficients, and both directions of the internal consistency calibration.

A The 85 free coordinates

Write the following ordered source-index list as $(s_0, \ldots, s_{84})$. The only independent unknowns of the principal equation are $y_0, \ldots, y_{84} \in \mathbb{Z}$, with $v_{s_i} = y_i$. In particular v_{219} is the joint quotient. All other coordinates that occur below are fixed polynomial abbreviations defined by the next appendix.

$$3, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18,$$
$$57, 99, 100, 101, 102, 103, 106, 107, 109, 110, 112, 113, 115, 116,$$
$$118, 119, 120, 122, 123, 127, 128, 131, 132, 133, 134, 138, 141, 142,$$
$$150, 151, 153, 154, 156, 157, 159, 160, 162, 163, 165, 166, 168, 169,$$
$$171, 172, 174, 175, 177, 178, 180, 181, 183, 184, 186, 187, 190, 191,$$
$$196, 197, 198, 199, 200, 205, 206, 207, 208, 209, 210, 211, 212, 218,$$
$$219.$$

B The complete integral reconstruction

For each table row (v, j, ϵ), remove the unique monomial ϵv_v from the literal source polynomial R_j in the next appendix. Call the resulting polynomial F_v. Define $v_v = -\epsilon F_v$, using previously defined coordinates. The table is in dependency order, not merely in increasing variable order. Thus these 124 instructions, together with the free list, are a complete division-free polynomial reconstruction. They are also a shorter finite specification than expanding every high-degree residual into monomials.

Table 3: Reconstruction instructions, read left-to-right and then down.

Step	v	Row j	ϵ	Step	v	Row j	ϵ
1	0	136	1	2	2	3	1
3	4	157	1	4	7	0	1
5	1	2	1	6	19	4	1
7	20	5	1	8	21	6	1
9	22	7	1	10	23	8	1
11	24	9	1	12	25	10	1
13	26	11	1	14	27	12	1
15	28	13	1	16	29	14	1
17	30	15	1	18	31	16	1
19	32	17	1	20	33	18	1
21	34	19	1	22	35	20	1
23	36	21	1	24	37	22	1
25	38	23	1	26	39	24	1
27	40	25	1	28	41	26	1
29	42	27	1	30	43	28	1
31	44	29	1	32	45	30	1
33	46	31	1	34	47	32	1
35	48	33	1	36	49	34	1
37	50	35	1	38	51	36	1
39	52	37	1	40	53	38	1
41	54	39	1	42	55	40	1

Step	v	Row j	ϵ	Step	v	Row j	ϵ
43	56	41	1	44	58	43	1
45	59	44	1	46	60	45	1
47	61	46	1	48	62	47	1
49	63	48	1	50	64	49	1
51	65	50	1	52	66	51	1
53	67	52	1	54	68	53	1
55	69	54	-1	56	70	55	-1
57	71	56	-1	58	72	57	-1
59	73	58	-1	60	74	59	-1
61	75	60	-1	62	76	61	-1
63	77	62	1	64	78	63	1
65	80	65	1	66	79	64	1
67	81	67	1	68	213	152	1
69	82	68	1	70	83	66	1
71	84	69	1	72	85	70	1
73	86	71	1	74	87	72	1
75	88	73	1	76	89	74	1
77	90	75	1	78	91	76	1
79	92	77	1	80	93	78	1
81	94	79	1	82	95	80	1
83	96	81	1	84	97	82	1
85	98	84	1	86	126	148	1
87	105	85	1	88	104	86	-1
89	129	93	1	90	108	87	-1
91	111	88	-1	92	114	89	-1
93	117	90	-1	94	121	91	-1
95	124	92	-1	96	125	94	1
97	130	95	1	98	135	96	-1
99	189	135	1	100	136	134	1
101	203	141	1	102	192	142	1
103	195	143	1	104	137	144	-1
105	139	97	1	106	140	98	1
107	143	100	1	108	152	108	1
109	155	110	1	110	158	112	1
111	161	114	1	112	164	116	1
113	167	118	1	114	170	120	1
115	173	122	1	116	176	124	1
117	179	126	1	118	182	128	1
119	185	130	1	120	188	132	1
121	193	145	1	122	194	146	1
123	201	137	1	124	202	139	1

C Complete retained source row table

The following 147 quadratic polynomials are the retained rows from the literal source219 system. Source labels use zero-based indices. Rows listed in the preceding pivot table define coordinates; the 23 rows in (10) contribute residual squares in (11). The additional joint residual is completely defined by (1)–(4) and Table 2. No other equations or unspecified constants enter the definition.

$$
\begin{aligned}
R_0 &= v_0 - v_0v_4 - v_4 + v_7\\
R_1 &= -1 + v_1 - v_2 - v_4v_5\\
R_2 &= -1 + v_1 - v_6v_7\\
R_3 &= 1 + v_2 - v_3v_{18} + 2v_{16}\\
R_4 &= -1 + 2v_8 - v_8^2 - 2v_8v_9 + 2v_9 - v_9^2 + v_{19}\\
R_5 &= -5 - 30v_8 - 36v_8^2 - 96v_8v_9 - 24v_8v_{19} - 40v_9 - 64v_9^2 - 32v_9v_{19}\\
&\quad - 10v_{19} - 4v_{19}^2 + v_{20}\\
R_6 &= -2 + 2v_8 - v_8^2 - 4v_8v_{20} + 4v_{20} - 4v_{20}^2 + v_{21}\\
R_7 &= 1 + 6v_8 - 36v_8^2 - 192v_8v_{20} - 24v_8v_{21} + 16v_{20} - 256v_{20}^2 - 64v_{20}v_{21}\\
&\quad + 2v_{21} - 4v_{21}^2 + v_{22}\\
R_8 &= -2 - v_9^2 - 2v_9v_{10} - v_{10}^2 + v_{23}\\
R_9 &= -9 + 7v_8 - v_8^2 - 4v_8v_9 - 4v_8v_{23} + 12v_9 - 4v_9^2 - 8v_9v_{23} + 12v_{23}\\
&\quad - 4v_{23}^2 + v_{24}\\
R_{10} &= -1 + 2v_8 - v_8^2 - 2v_8v_{10} + 2v_{10} - v_{10}^2 + v_{25}\\
R_{11} &= -6v_8 - 144v_8^2 - 192v_8v_9 - 192v_8v_{10} - 48v_8v_{19} - 48v_8v_{25} - 8v_9\\
&\quad - 64v_9^2 - 128v_9v_{10} - 32v_9v_{19} - 32v_9v_{25} - 64v_{10}^2 - 32v_{10}v_{19}\\
&\quad - 32v_{10}v_{25} - 2v_{19} - 4v_{19}^2 - 8v_{19}v_{25} - 4v_{25}^2 + v_{26}\\
R_{12} &= -4v_{24}^2 - 8v_{24}v_{26} - 4v_{26}^2 + v_{27}\\
R_{13} &= -6 + 16v_{10} - 16v_{10}^2 - 16v_{10}v_{23} - 16v_{10}v_{25} + 8v_{23} - 4v_{23}^2 - 8v_{23}v_{25}\\
&\quad + 8v_{25} - 4v_{25}^2 + v_{28}\\
R_{14} &= 1 + 6v_8 - 36v_8^2 - 96v_8v_9 - 48v_8v_{10} - 24v_8v_{19} - 48v_8v_{25}\\
&\quad - 24v_8v_{28} + 8v_9 - 64v_9^2 - 64v_9v_{10} - 32v_9v_{19} - 64v_9v_{25}\\
&\quad - 32v_9v_{28} + 8v_{10} - 16v_{10}^2 - 16v_{10}v_{19} - 32v_{10}v_{25} - 16v_{10}v_{28} + 2v_{19}\\
&\quad - 4v_{19}^2 - 16v_{19}v_{25} - 8v_{19}v_{28} + 8v_{25} - 16v_{25}^2 - 16v_{25}v_{28} + 4v_{28}\\
&\quad - 4v_{28}^2 + v_{29}\\
R_{15} &= -10 - 5v_8 - v_8^2 - 12v_8v_9 - 16v_8v_{10} - 4v_8v_{23} - 36v_9 - 36v_9^2\\
&\quad - 96v_9v_{10} - 24v_9v_{23} - 48v_{10} - 64v_{10}^2 - 32v_{10}v_{23} - 12v_{23} - 4v_{23}^2\\
&\quad + v_{30}\\
R_{16} &= -2 - 24v_8 - 36v_8^2 - 12v_8v_9 - 96v_8v_{10} - 24v_8v_{25} - 3v_9 - v_9^2\\
&\quad - 16v_9v_{10} - 4v_9v_{25} - 32v_{10} - 64v_{10}^2 - 32v_{10}v_{25} - 8v_{25} - 4v_{25}^2 + v_{31}\\
R_{17} &= 2 - 4v_{30}^2 - 8v_{30}v_{31} - 4v_{31}^2 + v_{32}
\end{aligned}
$$

$$
\begin{aligned}
R_{18} &= -12 + 16v_8 - 4v_8^2 - 8v_8v_{10} - 8v_8v_{19} - 8v_8v_{23} + 16v_{10} - 4v_{10}^2 \\
&\quad - 8v_{10}v_{19} - 8v_{10}v_{23} + 16v_{19} - 4v_{19}^2 - 8v_{19}v_{23} + 16v_{23} - 4v_{23}^2 + v_{33} \\
R_{19} &= -3 - 24v_8 - 64v_8^2 - 160v_8v_{10} - 64v_8v_{23} - 32v_8v_{25} - 32v_8v_{33} \\
&\quad - 26v_{10} - 100v_{10}^2 - 80v_{10}v_{23} - 40v_{10}v_{25} - 40v_{10}v_{33} - 8v_{23} - 16v_{23}^2 \\
&\quad - 16v_{23}v_{25} - 16v_{23}v_{33} - 6v_{25} - 4v_{25}^2 - 8v_{25}v_{33} - 4v_{33} - 4v_{33}^2 + v_{34} \\
R_{20} &= -1 - 18v_8 - 36v_8^2 - 96v_8v_9 - 24v_8v_{19} - 24v_9 - 64v_9^2 - 32v_9v_{19} \\
&\quad - 6v_{19} - 4v_{19}^2 + v_{35} \\
R_{21} &= -2 + 2v_8 - v_8^2 - 4v_8v_{35} + 4v_{35} - 4v_{35}^2 + v_{36} \\
R_{22} &= 1 + 6v_8 - 36v_8^2 - 192v_8v_{35} - 24v_8v_{36} + 16v_{35} - 256v_{35}^2 - 64v_{35}v_{36} \\
&\quad + 2v_{36} - 4v_{36}^2 + v_{37} \\
R_{23} &= -v_9^2 - 4v_9v_{37} - 4v_{37}^2 + v_{38} \\
R_{24} &= -6 - 30v_8 - 64v_8^2 - 128v_8v_9 - 64v_8v_{19} - 32v_9 - 64v_9^2 - 64v_9v_{19} \\
&\quad - 14v_{19} - 16v_{19}^2 + v_{39} \\
R_{25} &= -4 + 8v_9 - 64v_9^2 - 96v_9v_{10} - 32v_9v_{23} - 32v_9v_{39} + 6v_{10} - 36v_{10}^2 \\
&\quad - 24v_{10}v_{23} - 24v_{10}v_{39} + 2v_{23} - 4v_{23}^2 - 8v_{23}v_{39} + 4v_{39} - 4v_{39}^2 + v_{40} \\
R_{26} &= 8v_8 - 64v_8^2 - 96v_8v_{10} - 32v_8v_{25} - 32v_8v_{40} + 6v_{10} - 36v_{10}^2 \\
&\quad - 24v_{10}v_{25} - 24v_{10}v_{40} + 2v_{25} - 4v_{25}^2 - 8v_{25}v_{40} + 4v_{40} - 4v_{40}^2 + v_{41} \\
R_{27} &= -v_{10}^2 - 2v_{10}v_{11} - v_{11}^2 + v_{42} \\
R_{28} &= 2v_8 - v_8^2 - 4v_8v_{10} - 4v_8v_{42} + 4v_{10} - 4v_{10}^2 - 8v_{10}v_{42} + 4v_{42} \\
&\quad - 4v_{42}^2 + v_{43} \\
R_{29} &= -35 + 24v_8 - 4v_8^2 - 8v_8v_9 - 8v_8v_{23} - 8v_8v_{25} + 24v_9 - 4v_9^2 \\
&\quad - 8v_9v_{23} - 8v_9v_{25} + 24v_{23} - 4v_{23}^2 - 8v_{23}v_{25} + 24v_{25} - 4v_{25}^2 + v_{44} \\
R_{30} &= -16v_8^2 - 32v_8v_{10} - 32v_8v_{25} - 32v_8v_{42} - 16v_8v_{43} - 16v_8v_{44} \\
&\quad - 4v_{10} - 16v_{10}^2 - 32v_{10}v_{25} - 32v_{10}v_{42} - 16v_{10}v_{43} - 16v_{10}v_{44} \\
&\quad - 16v_{25}^2 - 32v_{25}v_{42} - 16v_{25}v_{43} - 16v_{25}v_{44} - 4v_{42} - 16v_{42}^2 \\
&\quad - 16v_{42}v_{43} - 16v_{42}v_{44} - 2v_{43} - 4v_{43}^2 - 8v_{43}v_{44} - 4v_{44}^2 + v_{45} \\
R_{31} &= v_{10} - 25v_{10}^2 - 40v_{10}v_{25} - 20v_{10}v_{28} - 16v_{25}^2 - 16v_{25}v_{28} - 4v_{28}^2 \\
&\quad + v_{46} \\
R_{32} &= -1 + 8v_9 - 4v_9^2 - 8v_9v_{23} + 8v_{23} - 4v_{23}^2 + v_{47} \\
R_{33} &= -1 - 16v_9^2 - 8v_9v_{10} - 32v_9v_{23} - 16v_9v_{47} - v_{10}^2 - 8v_{10}v_{23} \\
&\quad - 4v_{10}v_{47} - 16v_{23}^2 - 16v_{23}v_{47} - 4v_{47}^2 + v_{48} \\
R_{34} &= 1 + v_9 - v_9^2 - 4v_9v_{10} - 4v_9v_{48} - 4v_{10}^2 - 8v_{10}v_{48} - 4v_{48}^2 + v_{49} \\
R_{35} &= -1 - 4v_{49} - 4v_{49}^2 + v_{50} \\
R_{36} &= -1 - v_{10}^2 - 8v_{10}v_{49} - 4v_{10}v_{50} - 16v_{49}^2 - 16v_{49}v_{50} - 4v_{50}^2 + v_{51} \\
R_{37} &= 2 + v_9 - v_9^2 - 4v_9v_{10} - 4v_9v_{51} - 4v_{10}^2 - 8v_{10}v_{51} - 4v_{51}^2 + v_{52} \\
R_{38} &= -4v_{46}^2 - 8v_{46}v_{52} - 4v_{52}^2 + v_{53}
\end{aligned}
$$

$$R_{39} = -9 + 7v_8 - v_8^2 - 8v_8v_{46} - 4v_8v_{53} + 24v_{46} - 16v_{46}^2 - 16v_{46}v_{53} + 12v_{53} - 4v_{53}^2 + v_{54}$$

$$R_{40} = -1 - 4v_{54} - 4v_{54}^2 + v_{55}$$

$$R_{41} = -v_{10}^2 - 8v_{10}v_{54} - 4v_{10}v_{55} - 16v_{54}^2 - 16v_{54}v_{55} - 4v_{55}^2 + v_{56}$$

$$R_{42} = -v_8 - v_8^2 - 4v_8v_{10} - 4v_8v_{56} - 4v_{10} - 4v_{10}^2 - 8v_{10}v_{56} - 4v_{56} - 4v_{56}^2 + v_{57}$$

$$R_{43} = -10 - 12v_{57} - 4v_{57}^2 + v_{58}$$

$$R_{44} = 1 + 4v_9 - 4v_9^2 - 8v_9v_{23} - 16v_9v_{57} - 8v_9v_{58} + 4v_{23} - 4v_{23}^2 - 16v_{23}v_{57} - 8v_{23}v_{58} + 8v_{57} - 16v_{57}^2 - 16v_{57}v_{58} + 4v_{58} - 4v_{58}^2 + v_{59}$$

$$R_{45} = v_9 - 25v_9^2 - 40v_9v_{23} - 20v_9v_{59} - 16v_{23}^2 - 16v_{23}v_{59} - 4v_{59}^2 + v_{60}$$

$$R_{46} = -10 - 12v_{52} - 4v_{52}^2 + v_{61}$$

$$R_{47} = 1 - 16v_{52}^2 - 16v_{52}v_{60} - 16v_{52}v_{61} - 4v_{60}^2 - 8v_{60}v_{61} - 4v_{61}^2 + v_{62}$$

$$R_{48} = v_{10} - v_{10}^2 - 8v_{10}v_{60} - 4v_{10}v_{62} - 16v_{60}^2 - 16v_{60}v_{62} - 4v_{62}^2 + v_{63}$$

$$R_{49} = -4v_8 - 4v_8^2 - 4v_8v_{10} - 8v_8v_{19} - v_{10} - v_{10}^2 - 4v_{10}v_{19} - 4v_{19} - 4v_{19}^2 + v_{64}$$

$$R_{50} = 1 + 4v_8 - 4v_8^2 - 8v_8v_{19} - 8v_8v_{64} + 4v_{19} - 4v_{19}^2 - 8v_{19}v_{64} + 4v_{64} - 4v_{64}^2 + v_{65}$$

$$R_{51} = -2 - 4v_8 - 16v_8^2 - 32v_8v_{19} - 16v_8v_{63} - 16v_8v_{65} - 4v_{19} - 16v_{19}^2 - 16v_{19}v_{63} - 16v_{19}v_{65} - 4v_{63}^2 - 8v_{63}v_{65} - 2v_{65} - 4v_{65}^2 + v_{66}$$

$$R_{52} = -2 - 4v_{66} - 4v_{66}^2 + v_{67}$$

$$R_{53} = -v_2v_{13} + v_{68}$$

$$R_{54} = v_{12} - v_{13} - 2v_{22}v_{68} - v_{69}$$

$$R_{55} = v_{12} - v_{13} - 4v_{24}v_{68} - 2v_{27}v_{68} - v_{70}$$

$$R_{56} = v_{12} - v_{13} - 2v_{29}v_{68} - 6v_{68} - v_{71}$$

$$R_{57} = v_{12} - v_{13} - 4v_{30}v_{68} - 2v_{32}v_{68} - v_{72}$$

$$R_{58} = v_{12} - v_{13} - 2v_{34}v_{68} + 6v_{68} - v_{73}$$

$$R_{59} = v_{12} - v_{13} - 4v_{37}v_{68} - 2v_{38}v_{68} - v_{74}$$

$$R_{60} = v_{12} - v_{13} - 2v_{41}v_{68} - v_{75}$$

$$R_{61} = v_{12} - v_{13} - 2v_{45}v_{68} - 4v_{68} - v_{76}$$

$$R_{62} = -v_8v_9 + v_8v_{10} + v_9 + v_9v_{10} - v_{10} - v_{10}^2 + v_{77}$$

$$R_{63} = 1 + 2v_{11} + v_{11}^2 - v_{77}^2 + v_{78}$$

$$R_{64} = -v_{12}^2 + 2v_{12}v_{13} + 2v_{12}v_{80} - v_{13}^2 - 2v_{13}v_{80} - v_{78}^2 + v_{79} - v_{80}^2$$

$$R_{65} = -4v_{66}v_{68} - 2v_{67}v_{68} + 2v_{68} + v_{80}$$

$$R_{66} = -v_9v_{68} - v_{13} + v_{81}v_{82} + v_{83}$$

$$R_{67} = -1 + 2v_2 - 2v_2v_8 - 2v_2v_{19} + v_{81}$$

$$R_{68} = -v_8v_{213} - v_{12} + v_{82} + v_{213}$$

$$R_{69} = -6v_8v_{68} - 8v_9v_{68} + v_9v_{213} + v_{12} - v_{13} - 2v_{19}v_{68} + v_{84}$$

$$
\begin{aligned}
R_{70} &= -v_{12}v_{69} + v_{13}v_{69} + v_{85} \\
R_{71} &= -v_{70}v_{85} + v_{86} \\
R_{72} &= -v_{71}v_{86} + v_{87} \\
R_{73} &= -v_{72}v_{87} + v_{88} \\
R_{74} &= -v_{73}v_{88} + v_{89} \\
R_{75} &= -v_{74}v_{89} + v_{90} \\
R_{76} &= -v_{75}v_{90} + v_{91} \\
R_{77} &= -v_{76}v_{91} + v_{92} \\
R_{78} &= -v_{83}v_{92} + v_{93} \\
R_{79} &= -v_{84}v_{93} + v_{94} \\
R_{80} &= -v_{94}^2 + v_{95} \\
R_{81} &= -v_4 - 2v_4v_{16} + 2v_{16} + v_{96} \\
R_{82} &= -v_{14}v_{96} + v_{97} \\
R_{83} &= -2v_1^2 + 2v_1v_{12} + 2v_1v_{13} + 2v_1v_{97} + 2v_1v_{98} - v_{12}^2 - 2v_{12}v_{97} \\
&\quad - v_{13}^2 - 2v_{13}v_{98} + v_{17}v_{99} - v_{79}v_{95} - v_{97}^2 - v_{98}^2 \\
R_{84} &= -v_4v_{15} + v_{15} - v_{15}v_{96} + v_{98} \\
R_{85} &= -v_{100}v_{103} + v_{103} + v_{105} \\
R_{86} &= -v_3v_{126} + v_{101}v_{205} - v_{104} - v_{105}v_{126} \\
R_{87} &= v_{106}v_{205} - v_{107}v_{129} - v_{108} \\
R_{88} &= v_{109}v_{205} - v_{110}v_{129} - v_{111} \\
R_{89} &= v_{112}v_{205} - v_{113}v_{129} - v_{114} \\
R_{90} &= v_{115}v_{205} - v_{116}v_{129} - v_{117} \\
R_{91} &= v_{118}v_{206} - v_{120}v_{129} - v_{121} \\
R_{92} &= v_{122}v_{206} - v_{123}v_{129} - v_{124} \\
R_{93} &= -v_{17}v_{126} + v_{129} \\
R_{94} &= v_{125} - v_{127}v_{129} - v_{128} \\
R_{95} &= -v_{129}v_{131} + v_{130} - v_{132} \\
R_{96} &= -v_{129}v_{134} + v_{133}v_{205} - v_{135} \\
R_{97} &= v_{136} - 2v_{136}v_{138} + v_{139} \\
R_{98} &= -v_{136}v_{139} - v_{139} + v_{140} \\
R_{99} &= 2v_{137} + v_{137}^2 + v_{139}^2 - v_{140}^2 \\
R_{100} &= -v_{137}v_{142} - v_{142} + v_{143} \\
R_{101} &= -1 + v_{141}^2 + 2v_{141}v_{143} + v_{142}^2 \\
R_{108} &= -v_{137}v_{151} - v_{151} + v_{152} \\
R_{109} &= -1 + v_{150}^2 + 2v_{150}v_{152} + v_{151}^2 \\
R_{110} &= 1 + v_{137} - v_{137}v_{154} - v_{154} + v_{155} \\
R_{111} &= v_{153}^2 + 2v_{153}v_{155} - 2v_{154} + v_{154}^2
\end{aligned}
$$

$$R_{112} = -v_{137}v_{157} - v_{157} + v_{158}$$
$$R_{113} = -1 + v_{156}^2 + 2v_{156}v_{158} + v_{157}^2$$
$$R_{114} = -v_{137}v_{160} - v_{160} + v_{161}$$
$$R_{115} = -1 + v_{159}^2 + 2v_{159}v_{161} + v_{160}^2$$
$$R_{116} = -v_{137}v_{163} - v_{163} + v_{164}$$
$$R_{117} = -1 + v_{162}^2 + 2v_{162}v_{164} + v_{163}^2$$
$$R_{118} = -v_{137}v_{166} - v_{166} + v_{167}$$
$$R_{119} = -1 + v_{165}^2 + 2v_{165}v_{167} + v_{166}^2$$
$$R_{120} = -v_{137}v_{169} - v_{169} + v_{170}$$
$$R_{121} = -1 + v_{168}^2 + 2v_{168}v_{170} + v_{169}^2$$
$$R_{122} = -v_{137}v_{172} - v_{172} + v_{173}$$
$$R_{123} = -1 + v_{171}^2 + 2v_{171}v_{173} + v_{172}^2$$
$$R_{124} = -v_{137}v_{175} - v_{175} + v_{176}$$
$$R_{125} = -1 + v_{174}^2 + 2v_{174}v_{176} + v_{175}^2$$
$$R_{126} = 1 - v_{137}v_{178} - v_{178} + v_{179}$$
$$R_{127} = -1 + 2v_{177} + v_{177}^2 + 2v_{177}v_{179} + v_{178}^2$$
$$R_{128} = -v_{137}v_{181} - v_{181} + v_{182}$$
$$R_{129} = -1 + v_{180}^2 + 2v_{180}v_{182} + v_{181}^2$$
$$R_{130} = -v_{137}v_{184} - v_{184} + v_{185}$$
$$R_{131} = -1 + v_{183}^2 + 2v_{183}v_{185} + v_{184}^2$$
$$R_{132} = -v_{137}v_{187} - v_{187} + v_{188}$$
$$R_{133} = -1 + v_{186}^2 + 2v_{186}v_{188} + v_{187}^2$$
$$\begin{aligned} R_{134} = {} & -5 - 2v_0 - 1099511627780v_0^2 - 2v_0v_2 - v_1^2 - 2v_2^2 - v_3^2 - v_4^2 + 2v_8 \\ & - v_8^2 - v_9^2 - v_{10}^2 - v_{11}^2 - v_{12}^2 - v_{13}^2 - v_{14}^2 - v_{15}^2 - 4v_{16}^2 - v_{17}^2 \\ & + 2v_{100} - 2v_{100}^2 - v_{101}^2 - v_{102}^2 - v_{106}^2 - v_{109}^2 - v_{112}^2 - v_{115}^2 - v_{118}^2 - v_{119}^2 \\ & - v_{122}^2 - v_{125}^2 - v_{126}^2 - v_{130}^2 - v_{133}^2 + v_{136} - v_{189}^2 \end{aligned}$$
$$\begin{aligned} R_{135} = {} & 2v_3 - 4v_3v_8 - 4v_3v_9 - 4v_3v_{10} - 4v_3v_{11} - 4v_3v_{12} - 4v_3v_{13} \\ & - 4v_3v_{14} - 4v_3v_{15} - 8v_3v_{16} + v_{189} \end{aligned}$$
$$R_{136} = -16 + v_0 - 4v_{190}^2$$
$$R_{137} = -v_{137}v_{192} - v_{192} + v_{201}$$
$$R_{138} = -1 + v_{191}^2 + v_{192}^2 - v_{201}^2$$
$$R_{139} = -v_{194} - v_{194}v_{195} + v_{202}$$
$$R_{140} = -1 + v_{193}^2 + v_{194}^2 - v_{202}^2$$
$$R_{141} = -v_{184}^2 + v_{203}$$
$$R_{142} = v_{192} - 2v_{197}v_{203} - v_{203}$$
$$R_{143} = -4v_{192} - 8v_{192}v_{196} + v_{195}$$
$$R_{144} = -v_{137} - v_{191}v_{198} + v_{195}$$

$$
\begin{aligned}
R_{145} &= -v_{183} - v_{185} - v_{191}v_{199} + v_{193} \\
R_{146} &= -2v_{16} - 4v_{184}v_{200} + v_{194} \\
R_{148} &= -v_{102}v_{205} + v_{126} \\
R_{149} &= -v_{119}v_{206} + v_{126} \\
R_{150} &= -2 - 4v_{16}^2 - v_{142}^2 - v_{151}^2 + 2v_{154} - v_{154}^2 - v_{157}^2 - v_{160}^2 - v_{163}^2 - v_{166}^2 \\
&\quad - v_{169}^2 - v_{172}^2 - v_{175}^2 - v_{178}^2 - v_{181}^2 + v_{184}^2 - v_{187}^2 - v_{207}^2 - v_{208}^2 - v_{209} \\
&\quad - v_{209}^2 \\
R_{151} &= -1 - 4v_{104}^2 - 4v_{108}^2 - 4v_{111}^2 - 4v_{114}^2 - 4v_{117}^2 - 4v_{121}^2 - 4v_{124}^2 + v_{126}^2 \\
&\quad - 4v_{128}^2 - 4v_{132}^2 - 4v_{135}^2 - v_{210}^2 - v_{211}^2 - v_{212} - v_{212}^2 \\
R_{152} &= -v_2v_{12} + v_{213} \\
R_{157} &= -1 - v_2 - 2v_2v_{218} + v_4
\end{aligned}
$$

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